UNIVERSITY OF PERADENIYA - FACULTY OF SCIENCE
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER I) – 2023/2024
CSC1013/CS101 – INTRODUCTION TO COMPUTER SCIENCE AND PROGRAMMING)

Answer ALL Questions Time allowed: Two Hours

Instructions

- This is a closed book examination.
- Calculators are not allowed.
- There are Four (04) questions on four (04) pages.
- All questions carry equal marks.
- Clearly write the Question number and Part number in your answers.
- Make your answers concise and precise.

1. (a) Explain how the discipline of computer science differs from information technology. [03 marks]

   (b) Select the most appropriate answer. Write only the question number and answer letter in the answer book. Do not attach the question paper. [22 marks]

   i. How many bits are there in 2 Kilobytes (KB)?
   (a) 1,024 (b) 8,192 (c) 16,384 (d) 2,048

   ii. A 600 MB file is downloaded over a 20 Mbps connection. Theoretically how long will it take?
   (a) 3 minutes (b) 4 minutes (c) 5 minutes (d) 6 minutes

   iii. What is the decimal equivalent of the binary number 101101_2?
   (a) 45 (b) 41 (c) 37 (d) 33

   iv. What is the hexadecimal equivalent of the decimal number 254?
   (a) 1FE (b) FE (c) F4 (d) EF

   v. Which logic gate produces output 1 only when all inputs are 0?
   (a) AND (b) NAND (c) NOR (d) XOR

   vi. What is the result of BCD addition for decimal $47 + 38$?
   (a) 1000 0101 (b) 0111 1111 (c) 1001 0001 (d) 1010 0110

   vii. Using a Caesar cipher with a shift of 3, what is the encrypted form of the word "SECRET"?
   (a) VHFUHW (b) VHFUHT (c) VJHFUX (d) UHFQEW

   viii. Which of the following is a combinational circuit?
   (a) Flip-flop (b) Counter (c) Multiplexer (d) Latch

   ix. What decimal value does 11110110 represent in $\langle 8, 3\rangle$ signed fixed-point format?
   (a) -1.5 (b) -2.625 (c) -10.5 (d) -1.25

x. What is the function of a DNS server?

A. Encrypt data packets for secure transmission.
B. Store user passwords securely.
C. Host web applications.
D. Translate domain names into IP addresses.

xi. Which of the following best describes the Agile methodology?

A. Linear, step-by-step approach.
B. Process with heavy documentation before coding.
C. Iterative development with regular stakeholder feedback.
D. Code-first, test-later process.

2. (a) Convert the decimal number -10.625 into 32-bit IEEE-754 floating-point format. [04 marks]

(b) Perform the following binary arithmetic operations. Clearly indicate the steps. *(No marks will be given if steps are not indicated, or operation is performed by converting to decimal values).*

i. Add the binary numbers 1011 and 1101. [02 marks]

ii. $00101101 + 00001111$ [04 marks]

(c) Simplify the Boolean expression using a Karnaugh Map:

$$F(A, B, C, D) = \Sigma(0, 2, 5, 7, 8, 10, 13, 15)$$

[04 marks]

(d) Express the function $F = (A + B)(A' + C)$ in both SOP and POS forms. [04 marks]

(e) Simplify the following Boolean expression:

$$F(A, B, C) = AB + A'B + AB'C$$

Provide the step-by-step simplification using Boolean algebra. [04 marks]

(f) Draw the simplified circuit in (e) using NAND and NOR gates. *HINT: use De Morgan's theorem.* [03 marks]

3. The University of Peradeniya is looking for a programmer to develop a complete system for accurately tracking student attendance. Manual attendance in lecture halls often leads to errors, proxy attendance, and record-keeping delays. To address these issues, the university plans to implement a digital attendance management system for easier monitoring and reporting.

   The proposed system should be able to:

   (a) Allow lecturers to take attendance digitally (via app or web portal).
   (b) Automatically update the student's attendance record in the university database.
   (c) Send alerts to students and faculty when attendance drops below 75%.
   (d) Allow students to view their attendance records at any time.
   (e) Generate attendance reports for departments at the end of each semester.

   Assume you are a member of the programming group who is going to take this task to develop the system. Based on the above scenario, answer the following questions.

   (a) Draw a flowchart using standard symbols representing the process of making attendance for a class (E.g., A lecture of CSC1013). This function should have the following requirements: [10 marks]
      i. The lecturer selects a course.
      ii. The system will load all registered students into the program.
      iii. The lecturer marks each student as Present or Absent.
      iv. Attendance is saved to the database.
      v. End after all students are processed.

   (b) Convert the flowchart designed in part a to a pseudo-code. [05 marks]

   (c) You have decided to use C as the programming language as a team. Define a suitable structure for the above system to handle students' data within the program. [05 marks]

   (d) The following code snippet is part of the attendance management system for inputting student attendance. During execution, the program encounters runtime error/s ("stack overflow").

```c
#include <stdio.h>

void takeAttendance(int studentIDs[]) {
    int count = 0;
    while (1) {
        int currentID = studentIDs[count];
        printf("Taking attendance for ID: %d\n", currentID);
        count = count + 1;
    }
}

int main() {
    int ids[] = {101, 102, 103, 104};
    takeAttendance(ids);
    return 0;
}
```

      [i.]

      i. Identify the error/s and explain why the error/s cause a stack overflow. [02 marks]
      ii. Rewrite this code after correcting the error. [03 marks]

4. (a) What are the differences between lists, tuples, and sets in Python? Explain your answer with real-world examples. [05 marks]

   (b) Write the outputs of the following Python code snippets. [08 marks]

      i.

```
x = 10
def func():
    x = 20
    print(x, end=" ")
func()
print(x)
```

      ii.

```
numbers = [1, 2, 3, 4]
squares = [n**2 for n in numbers if n % 2 == 0]
print(squares)
```

      iii.

```
def append_to_list(value, my_list=[]):
    my_list.append(value)
    return my_list

print(append_to_list(1))
print(append_to_list(2))
```

      iv.

```
numbers = (1, 2, 3, 4)
numbers[1] = 10
print("Modified Tuple: ", numbers)
```

   (c) Write a Python program for the function `checkPassword` to check if the length of a given password string is 10 characters or not. If not, deduct 5 points per missing character. If the total deduction is greater than 30 points, print out the deduction and message "The password is unsafe! Please reset."; otherwise, print out "The password is safe." [12 marks]